

- 4** Mrs Tan bought  $x$  kg of rice for \$65 in December 2019. In February 2020, the price of rice increased and she received 6 kg less for the same amount of money spent.

**(a)** Write down an expression for the price of rice per kilogram, in terms of  $x$ ,

**(i)** in December 2019,

*Answer* \$..... [1]

**(ii)** in February 2020.

*Answer* \$..... [1]

**(b)** If the increase in price is \$2 per kilogram of rice, form an equation in  $x$  and show that it reduces to  $x^2 - 6x - 195 = 0$ .

*Answer*

[3]

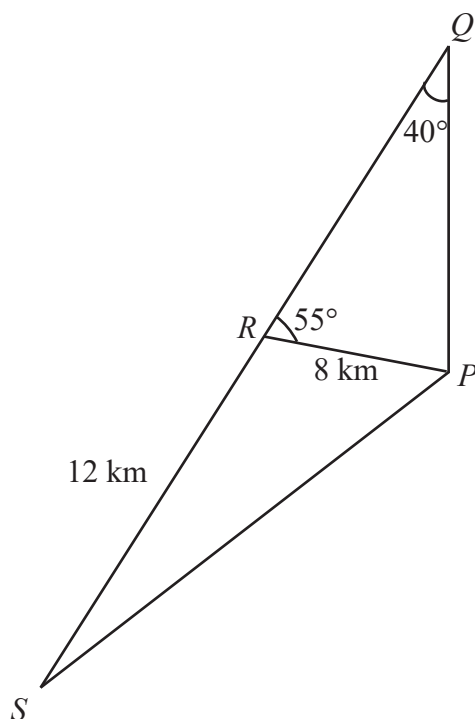
**(c)** Solve the equation, giving your answers correct to 2 decimal places.

*Answer*  $x = \dots\dots\dots$  or  $\dots\dots\dots$  [2]

**(d)** Hence, find the price of rice per kilogram in February 2020, leaving your answer to the nearest cent.

*Answer* ..... [1]

- 5 In the diagram,  $PQRS$  are four points on level ground, and  $Q$  is due north of  $P$ .



Given that angle  $PQR = 40^\circ$ , angle  $PRQ = 55^\circ$ ,  $PR = 8$  km,  $RS = 12$  km and that  $QRS$  is a straight line,

- (a) show that the distance  $PS$  is 17.836 km.

*Answer*

[3]

- (b) Using the result in (a), find,  
 (i) the bearing of  $P$  from  $S$ ,

o

- Answer* ..... [3]  
 (ii) the shortest distance from  $R$  to  $PS$ .

- Answer* .....km [2]  
 (c) A vertical tower stands at point  $R$ . Evan, walking along  $PS$  and stops at a point where the greatest angle of elevation of the top of the tower,  $T$ , is  $3^\circ$ . Find the height of the tower.

*Answer* .....km [2]